

# PNS – Surp Quiz 2

## Detailed Solutions and Marking Rubric

### Question 1 – Normal Distribution

Let

$$X \sim N(0, 1)$$

denote a standardized measurement error, and let

$$C = \{X > 0\}.$$

We solve each part separately.

**(a) Find  $\mathbb{P}(C)$  using the standard normal CDF  $\Phi(\cdot)$ .**

We are given

$$C = \{X > 0\}.$$

Therefore,

$$\mathbb{P}(C) = \mathbb{P}(X > 0).$$

Using the complement rule,

$$\mathbb{P}(X > 0) = 1 - \mathbb{P}(X \leq 0).$$

Since  $X \sim N(0, 1)$ , its CDF is the standard normal CDF  $\Phi$ . Hence,

$$\mathbb{P}(X \leq 0) = \Phi(0).$$

By symmetry of the standard normal distribution,

$$\Phi(0) = \frac{1}{2}.$$

Therefore,

$$\boxed{\mathbb{P}(C) = 1 - \Phi(0) = \frac{1}{2}}$$

Thus,

$$\boxed{\mathbb{P}(C) = 0.5}.$$

**(b) Find  $f_{X|C}(x)$  and state its domain. What is this distribution called?**

The conditional density of a continuous random variable given an event  $C$  with  $\mathbb{P}(C) > 0$  is

$$f_{X|C}(x) = \frac{f_X(x)\mathbf{1}_C(x)}{\mathbb{P}(C)}.$$

Here,

$$C = \{X > 0\}.$$

The standard normal density is

$$f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}, \quad x \in \mathbb{R}.$$

From part (a),

$$\mathbb{P}(C) = \frac{1}{2}.$$

Therefore,

$$f_{X|C}(x) = \frac{\frac{1}{\sqrt{2\pi}}e^{-x^2/2}}{\frac{1}{2}}, \quad x > 0.$$

Hence,

$$f_{X|C}(x) = \frac{2}{\sqrt{2\pi}}e^{-x^2/2}, \quad x > 0.$$

Equivalently, using the standard normal density  $\phi(x)$ ,

$$\boxed{f_{X|C}(x) = 2\phi(x), \quad x > 0.}$$

For  $x \leq 0$ , the conditional density is zero because the event  $C = \{X > 0\}$  restricts the possible values of  $X$  to the positive half-line.

Thus,

$$\boxed{f_{X|C}(x) = \begin{cases} 2\phi(x), & x > 0, \\ 0, & x \leq 0. \end{cases}}$$

The domain is therefore

$$\boxed{x > 0}.$$

This distribution is called the **\*\*half-normal distribution\*\*** (specifically, the positive half of a standard normal distribution).

$$\boxed{X \mid X > 0 \sim \text{Half-Normal}(0, 1)}$$

**(c) Using Bayes' rule for events, explain why  $f_{X|C}(x) = 0$  for  $x \leq 0$ .**

~~We want to understand why the conditional density must be zero for  $x \leq 0$ .~~

~~Consider a small interval around a value  $x \leq 0$ . Since~~

$$~~C = \{X > 0\},~~$$

~~the events~~

$$~~\{X \in dx\} \text{ and } C~~$$

~~cannot occur simultaneously when  $x \leq 0$ .~~

~~Using the conditional probability formula,~~

$$~~\mathbb{P}(X \in dx | C) = \frac{\mathbb{P}(X \in dx, C)}{\mathbb{P}(C)}.~~$$

~~For  $x \leq 0$ ,~~

$$~~\mathbb{P}(X \in dx, C) = 0,~~$$

~~because  $X \leq 0$  contradicts the event  $C = \{X > 0\}$ .~~

~~Therefore,~~

$$~~\mathbb{P}(X \in dx | C) = 0.~~$$

~~Consequently, the conditional density must satisfy~~

$$~~f_{X|C}(x) = 0, \quad x \leq 0.~~$$

~~More generally, the conditional density can be written as~~

$$~~f_{X|C}(x) = \frac{f_X(x) \mathbf{1}_{\{x > 0\}}}{\mathbb{P}(C)}.~~$$

~~The indicator function~~

$$~~\mathbf{1}_{\{x > 0\}} = \begin{cases} 1, & x > 0, \\ 0, & x \leq 0 \end{cases}~~$$

~~automatically makes the conditional density zero outside the event.~~

$$f_{X|Y}(x|y) = \frac{f_X(x)f_{Y|X}(y|x)}{f_Y(y)}.$$

can also write f(c)  
as p(c) and f(c  
given x) as p(c|x)

**(d) Find  $\mathbb{E}[X | C]$ .**

We have already established that

$$f_{X|C}(x) = 2\phi(x), \quad x > 0.$$

The definition of conditional expectation for a continuous random variable gives

$$\mathbb{E}[X | C] = \int_{-\infty}^{\infty} x f_{X|C}(x) dx.$$

Since the conditional density is zero for  $x \leq 0$ ,

$$\mathbb{E}[X \mid C] = \int_0^\infty x(2\phi(x)) dx.$$

Using

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

we obtain

$$\mathbb{E}[X \mid C] = \frac{2}{\sqrt{2\pi}} \int_0^\infty x e^{-x^2/2} dx.$$

Now use the substitution

$$u = \frac{x^2}{2}.$$

Then

$$du = x dx.$$

When  $x = 0$ ,  $u = 0$ , and as  $x \rightarrow \infty$ ,  $u \rightarrow \infty$ .

Therefore,

$$\mathbb{E}[X \mid C] = \frac{2}{\sqrt{2\pi}} \int_0^\infty e^{-u} du.$$

Since

$$\int_0^\infty e^{-u} du = 1,$$

we get

$$\mathbb{E}[X \mid C] = \frac{2}{\sqrt{2\pi}}.$$

Simplifying,

$$\boxed{\mathbb{E}[X \mid C] = \sqrt{\frac{2}{\pi}}}.$$

Numerically,

$$\boxed{\mathbb{E}[X \mid C] \approx 0.7979}.$$

This also agrees with the known result

$$\mathbb{E}[X \mid X \geq 0] = \sqrt{\frac{2}{\pi}}$$

for a standard normal random variable. Since the normal distribution is continuous,

$$\mathbb{P}(X = 0) = 0,$$

so conditioning on  $X > 0$  and  $X \geq 0$  gives the same expectation.

## Question 2 – Uniform Distribution

Let

$$X \sim \text{Uniform}(0, 10)$$

represent the arrival time, in minutes, of a bus.

Let

$$B = \{X > 4\}.$$

Thus, event  $B$  means that the bus has not arrived by minute 4.

The density of  $X$  is

$$f_X(x) = \begin{cases} \frac{1}{10}, & 0 < x < 10, \\ 0, & \text{otherwise.} \end{cases}$$

**(a) Find  $\mathbb{P}(B)$ .**

We have

$$B = \{X > 4\}.$$

Therefore,

$$\mathbb{P}(B) = \mathbb{P}(X > 4).$$

Since  $X$  is uniformly distributed on  $(0, 10)$ , the probability is proportional to the length of the corresponding interval.

The interval corresponding to  $X > 4$  is

$$(4, 10),$$

whose length is

$$10 - 4 = 6.$$

The total interval has length

$$10 - 0 = 10.$$

Hence,

$$\mathbb{P}(X > 4) = \frac{6}{10}.$$

Therefore,

$$\boxed{\mathbb{P}(B) = \frac{3}{5} = 0.6.}$$

Alternatively, using the CDF,

$$\mathbb{P}(X > 4) = 1 - \mathbb{P}(X \leq 4) = 1 - \frac{4}{10} = \frac{6}{10} = \frac{3}{5}.$$

**(b) Find  $f_{X|B}(x)$  and state its domain.**

The conditional density is

$$f_{X|B}(x) = \frac{f_X(x)\mathbf{1}_B(x)}{\mathbb{P}(B)}.$$

We know

$$f_X(x) = \frac{1}{10}, \quad 0 < x < 10,$$

and

$$\mathbb{P}(B) = \frac{3}{5}.$$

Since

$$B = \{X > 4\},$$

the conditional density is nonzero only when

$$x > 4.$$

Thus,

$$f_{X|B}(x) = \frac{\frac{1}{10}}{\frac{3}{5}}, \quad 4 < x < 10.$$

Simplifying,

$$f_{X|B}(x) = \frac{1}{10} \cdot \frac{5}{3} = \frac{1}{6}.$$

Therefore,

$$f_{X|B}(x) = \begin{cases} \frac{1}{6}, & 4 < x < 10, \\ 0, & \text{otherwise.} \end{cases}$$

The domain is

$$4 < x < 10.$$

Notice that the density integrates to 1:

$$\int_4^{10} \frac{1}{6} dx = \frac{1}{6}(10 - 4) = \frac{6}{6} = 1.$$

**(c) Comment on the shape of  $f_{X|B}(x)$ . Is it still uniform?**

Yes.

The conditional density is

$$f_{X|B}(x) = \frac{1}{6}, \quad 4 < x < 10.$$

This is a constant density over the interval  $(4, 10)$ .

Therefore,

$$X \mid B \sim \text{Uniform}(4, 10).$$

So the distribution is still uniform, but over a smaller interval.

The reason follows directly from the definition of conditional density:

$$f_{X|B}(x) = \frac{f_X(x)}{\mathbb{P}(B)}$$

for values of  $x$  satisfying the event  $B$ .

The original density was constant:

$$f_X(x) = \frac{1}{10}.$$

Conditioning on  $B = \{X > 4\}$  removes the portion of the original interval from 0 to 4.

The remaining interval is

$$(4, 10).$$

The density is then renormalized so that the total probability is 1:

$$\frac{1/10}{3/5} = \frac{1}{6}.$$

Thus, the shape remains flat, but the height increases from  $1/10$  to  $1/6$  because the probability mass is now concentrated entirely on the shorter interval  $(4, 10)$ .

Hence,

$$X \mid X > 4 \sim \text{Uniform}(4, 10).$$

**(d) Compute  $\mathbb{E}[X \mid B]$ .**

From part (b),

$$f_{X|B}(x) = \frac{1}{6}, \quad 4 < x < 10.$$

Therefore,

$$\mathbb{E}[X \mid B] = \int_{-\infty}^{\infty} x f_{X|B}(x) dx.$$

Since the conditional density is nonzero only on  $(4, 10)$ ,

$$\mathbb{E}[X \mid B] = \int_4^{10} x \frac{1}{6} dx.$$

Hence,

$$\mathbb{E}[X \mid B] = \frac{1}{6} \int_4^{10} x dx.$$

Using

$$\int x dx = \frac{x^2}{2},$$

we obtain

$$\mathbb{E}[X \mid B] = \frac{1}{6} \left[ \frac{x^2}{2} \right]_4^{10}.$$

Thus,

$$\mathbb{E}[X \mid B] = \frac{1}{6} \left( \frac{100}{2} - \frac{16}{2} \right).$$

Therefore,

$$\mathbb{E}[X \mid B] = \frac{1}{6}(50 - 8) = \frac{42}{6} = 7.$$

Hence,

$$\boxed{\mathbb{E}[X \mid B] = 7.}$$

This also follows immediately from the fact that

$$X \mid B \sim \text{Uniform}(4, 10),$$

and the mean of a  $\text{Uniform}(a, b)$  random variable is

$$\frac{a + b}{2}.$$

Therefore,

$$\mathbb{E}[X \mid B] = \frac{4 + 10}{2} = \boxed{7}.$$